

On-Shell Methods for Tree-Level Amplitudes in (De)Constructed Gauge Theory

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Contents

- 1 Motivation
- 2 Preliminary
- 3 Model and Computation
- 4 Some problems and extends
- 5 Summary

Why We Study Scattering Amplitudes?

1. Bridge between theory and experiment

- Core prediction targets for high-energy collider experiments such as the LHC , especially for high multiplicity amplitudes.
- Any new theory (SUSY, GUTs, extra dimensions) must predict observable cross sections

Why We Study Scattering Amplitudes?

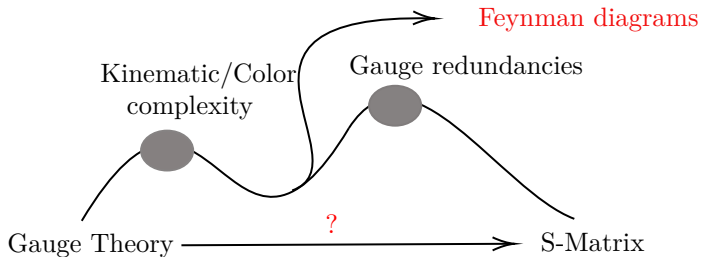
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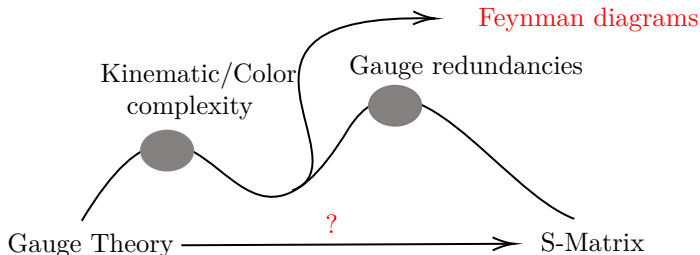
2. Reveal deep structures of quantum field theory

- Amplitudes exhibit hidden symmetries not visible in the Lagrangian
- These symmetries suggest deeper theoretical frameworks, such as amplituhedra or holographic principle.

Challenges we face before



Challenges we face before



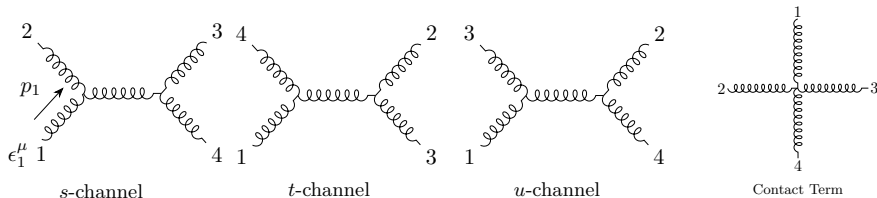
n pt. amplitudes	4	5	6	7	8	9	10
# of diagrams	4	25	220	2485	34300	559405	10525900

(tree-level gluon scattering)

The number of Feynman diagrams grow quite rapidly!

Conventional Computation

Usually, when we compute the gluon amplitudes by using Feynman diagram, we will obtain something like



$$\begin{aligned} \mathcal{M}_s(p_1 p_2 \rightarrow p_3 p_4) = & -\frac{g_s^2}{s} f^{abe} f^{cde} \\ & \times \left\{ -4 (\epsilon_1 \cdot \epsilon_3^*) (\epsilon_2 \cdot p_1) (\epsilon_4^* \cdot p_3) + 2 (\epsilon_1 \cdot \epsilon_2) (\epsilon_3^* \cdot p_1) (\epsilon_4^* \cdot p_3) \right. \\ & \left. - 2 (\epsilon_1 \cdot p_4) (\epsilon_2 \cdot p_1) (\epsilon_3^* \cdot \epsilon_4^*) + (\epsilon_1 \cdot \epsilon_2) (p_4 \cdot p_1) (\epsilon_3^* \cdot \epsilon_4^*) + 10 \text{ terms} \right\} \end{aligned}$$

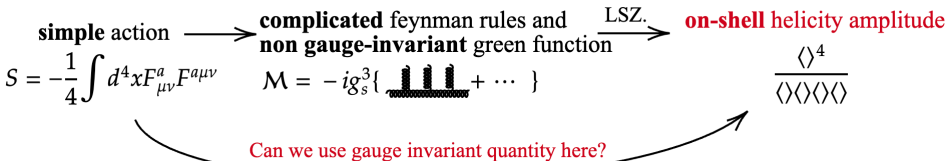
If you consider 5point case, it will become worse:

★ We have 25 diagrams and nearly 10000 terms!



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General Properties

- Lorentz covariance $\Rightarrow$ M as a function of Lorentz invariants $p_i \cdot p_j$, $p_i \cdot \epsilon_j$ and $\epsilon_i \cdot \epsilon_j$.
- Analyticity $\Rightarrow$ M is an analytic function of $p_i \cdot p_j$ and $p_i \cdot \epsilon_j$
- Unitarity $\Rightarrow$ simple poles $\leftrightarrow$ single particles
 $\Rightarrow$ branch cuts $\leftrightarrow$ continuous states
- Tree Level $\Rightarrow$ M is a rational function

Historically, when people consider 4-point scattering

$$p_1, p_2, p_3, p_4, \quad \text{satisfying} \quad p_i^2 = 0 \quad \text{and} \quad \sum_{i=1}^4 p_i = 0$$

$$\text{d.o.f} = 4n - 4 - n = 3n - 4 \quad \text{independent complexed d.o.f}$$

This seems to be too many, so people change to use Mandelstam variables (for $\pi\pi$ scattering)

$$s, t, u \quad \text{satisfying} \quad s + t + u = 0$$

Also for earliest String Amplitude (Veneziano Amplitude)

$$A_{\text{open}}(s, t) = g^2 \frac{\Gamma(-\alpha' s) \Gamma(-\alpha' t)}{\Gamma(-\alpha' s - \alpha' t)}$$

But the problem is that Mandelstam variables only apply to $2 \rightarrow 2$ scattering, how about higher point?

single complexed variable like $A(z)$

This is the starting point of BCFW recursion relation!

Poincaré Invariance and Little Group Scaling

We consider a Poincaré invariant field theory. The one-particle state are defined from a reference momentum k like this

$$|p; \sigma\rangle = U(L(p; k))|k; \sigma\rangle$$

Then the general Lorentz transformation acts like

$$\begin{aligned} U(\Lambda)|p, \sigma\rangle &= U(\Lambda)U(L(p; k))|k, \sigma\rangle \\ &= U(L(\Lambda p; k))U(L^{-1}(\Lambda p; k)\Lambda L(p; k))|k, \sigma\rangle \end{aligned}$$

here $W = L^{-1}(\Lambda p; k)\Lambda L(p; k)$ is not a general transformation but keep k invariant, called little group. Thus we have

$$U(W(\Lambda, p, k))|k; \sigma\rangle = D_{\sigma\sigma'}(W(\Lambda, p, k))|k, \sigma'\rangle$$

so one-particle state with momentum p transformed like

$$U(\Lambda)|p, \sigma\rangle = D_{\sigma\sigma'}(W(\Lambda, p, k))|\Lambda p, \sigma'\rangle$$

One-particle state transformed under the little group!

For massless particle, the little group is $SO(2) \simeq U(1)$, so the representation matrix is just a phase. The Poincaré invariance of S-matrix requires

$$\mathcal{M}^\Lambda(p_a, \sigma_a) = \prod_a (D_{\sigma\sigma'}) \mathcal{M}((\Lambda p)_a, \sigma'_a)$$

- **Massless Case**

$$p_\mu \sigma^\mu = p_{\alpha\dot{\alpha}} = \lambda_\alpha \tilde{\lambda}_{\dot{\alpha}} = |\lambda\rangle [\lambda|$$

There is an ambiguity for the definition, the momentum is invariant under the following redefinition

$$\lambda \rightarrow t^{-1}\lambda, \quad \tilde{\lambda} \rightarrow t\tilde{\lambda}, \quad t \in \mathbb{C}$$

same for

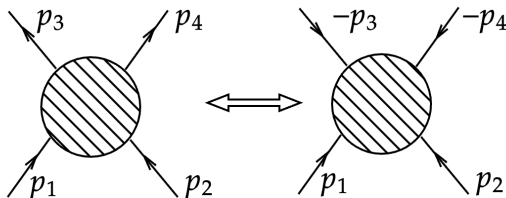
$$|\lambda\rangle \rightarrow t^{-1}|\lambda\rangle, \quad [\lambda] \rightarrow t[\lambda]$$

The scattering amplitudes should transform **covariantly** under little group scaling:

$$\mathcal{A}_n(\{|1\rangle, |1], h_1\}, \dots \{t_i^{-1}|i\rangle, t_i|i], h_i\}, \dots) = t_i^{2h_i} \mathcal{A}_n$$

Crossing Symmetry and MHV Classification

Crossing symmetry is a result from analyticity, unitarity and Lorentz invariance.



It means $M(p_1 p_2 \rightarrow p_3 p_4)$ can be transformed to $M(p_1 p_2 \leftarrow p_3 p_4)$ by analytic continuation.

Remember that the definition of helicity is $h = \frac{\vec{S} \cdot \vec{p}}{|\vec{p}|}$, so the sign will be flipped under crossing symmetry.

In this report, we choose the convention that all particles are **ingoing**.

The concept “**MHV** (maximally helicity violating)” comes from thinking a $2 \rightarrow n - 2$ scattering process.

By crossing symmetry, an outgoing gluon with $\left\{ \begin{array}{c} \text{negative} \\ \text{positive} \end{array} \right\}$ helicity is an incoming gluon with $\left\{ \begin{array}{c} \text{positive} \\ \text{negative} \end{array} \right\}$ helicity.

$$A[1^+ 2^+ 3^- 4^- 5^+ \cdots n^+] \quad \text{crossed over to} \quad 1^- 2^- \rightarrow 3^- 4^- 5^+ \cdots n^+$$

This is case in which we can maximally violate the helicity, so called **MHV amplitudes**.

NMHV(Next to MHV) can be similarly defined by including 3 minus helicity and $n-3$ plus helicity gluon, like

$$A[1^+ 2^+ 3^- 4^- 5^- 6^+ \cdots n^+]$$

So, N^K MHV described amplitudes with $K + 2$ negative helicities and $n - K - 2$ positive helicities.

Color-ordering for Yang-Mills

Consider the Yang-Mills Lagrangian

$$\mathcal{L} = -\frac{1}{2}\text{Tr}(F_{\mu\nu}F^{\mu\nu})$$

The 3 point and 4 point vertices include $\tilde{f}^{abc}$ and $\tilde{f}^{abe}\tilde{f}^{cde} + \text{perms.}$
(With a different convention, $\text{Tr}[T^a T^b] = \delta^{ab}$ and $[T^a, T^b] = i\tilde{f}^{abc}T^c$)

We have

$$c_s = \tilde{f}^{a_1 a_2 b} \tilde{f}^{b a_3 a_4}, \quad c_t = \tilde{f}^{a_4 a_1 b} \tilde{f}^{b a_2 a_3}, \quad c_u = \tilde{f}^{a_1 a_3 b} \tilde{f}^{b a_2 a_4}$$

and the color factor can be rewritten by the trace of product of generators

$$i\tilde{f}^{abc} = \text{Tr}([T^a, T^b]T^c),$$

Moreover, in $SU(N)$, we have a Fierz identity

$$\sum_a T_{ij}^a T_{kl}^a = \delta_{il} \delta_{kj} - \frac{1}{N} \delta_{ij} \delta_{kl}.$$

This identity is easier understood as matrix form like

$$\text{Tr}\{T^a A\}\text{Tr}\{T^a B\} = \text{Tr}\{AB\} - \frac{1}{N}\text{Tr}\{A\}\text{Tr}\{B\},$$

$$\text{Tr}\{AT^a BT^a\} = \text{Tr}\{A\}\text{Tr}\{B\} - \frac{1}{N}\text{Tr}\{AB\}.$$

So, the 4 gluon s-channel gives us

$$\begin{aligned}\tilde{f}^{a_1 a_2 b} \tilde{f}^{b a_3 a_4} &= \text{Tr}(T^{a_1} T^{a_2} T^{a_3} T^{a_4}) - \text{Tr}(T^{a_2} T^{a_1} T^{a_3} T^{a_4}) \\ &\quad - \text{Tr}(T^{a_1} T^{a_2} T^{a_4} T^{a_3}) + \text{Tr}(T^{a_2} T^{a_1} T^{a_4} T^{a_3}).\end{aligned}$$

Therefore, the full 4-point amplitude can be rewritten like

$$\mathcal{A}_{4,\text{tree}} = g^2 (A_4[1234] \text{Tr}(T^{a_1} T^{a_2} T^{a_3} T^{a_4}) + \text{perms of } (234))$$

here the subamplitudes $A_4[1234]$, $A_4[1243]$, etc. are called **color-ordered amplitudes**. This concept can be easily generalized to tree-level n-point case

$$\mathcal{A}_{n,\text{tree}} = g^{n-2} \sum_{\sigma} A_n[1, \sigma(2, 3 \cdots n)] \text{Tr}(T^{a_1} T^{\sigma(a_2 \cdots a_n)})$$

Old Results from New Perspective

Parke–Taylor Formula (MHV amplitudes):

$$A_5[1^+2^+3^+4^+5^+] = 0$$

$$A_5[1^-2^+3^+4^+5^+] = 0$$

$$A_5[1^-2^-3^+4^+5^+] = \frac{\langle 12 \rangle^4}{\langle 12 \rangle \langle 23 \rangle \langle 34 \rangle \langle 45 \rangle \langle 51 \rangle} \quad (\text{first non-trivial one})$$

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$$A_5[1^+2^-3^-4^-5^-] = 0$$

$$A_5[1^+2^+3^-4^-5^-] = \frac{[12]^4}{[12][23][34][45][51]}$$

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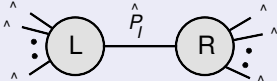
$$A_5[1^+2^+3^-4^-5^-] = \frac{[12]^4}{[12][23][34][45][51]}$$

This kind of computation can be reproduced by On-shell method.

Gauge Theory $\xrightarrow{\text{On-shell method}}$ S-Matrix

The Introduction to BCFW Recursion Relation

BCFW recursion relation

$$A_n = \sum_{\text{diagrams } I} \hat{A}_L(z_I) \frac{1}{P_I^2} \hat{A}_R(z_I) = \sum_{\text{diagrams } I} \hat{A}_L(z_I) \frac{1}{P_I^2} \hat{A}_R(z_I)$$


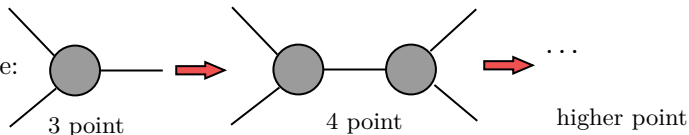
The diagram illustrates the BCFW recursion relation. It shows two vertices, labeled L and R, connected by a horizontal line representing a propagator. Above the propagator line is the label $\hat{P}_I$. Vertex L is on the left and has three external lines extending from it, each labeled with a hat symbol (^). Vertex R is on the right and also has three external lines extending from it, each labeled with a hat symbol (^). The entire equation and diagram are enclosed in a light purple box.

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More understandable:



★ From lower point on-shell amp. to higher point on-shell amp.!!

Momentum Shift in BCFW

What did BCFW do to make the shift?

Here we consider the case in which all particles are massless, $p_i^2 = 0$ for all $i = 1, 2, \dots, n$. We choose two momentum to be shifted oppositely

$$\begin{aligned} p_i &\rightarrow \hat{p}_i(z) \equiv p_i - zk, & p_j &\rightarrow \hat{p}_j(z) \equiv p_j + zk \\ |\hat{i}\rangle &= |i\rangle - z|j\rangle, & |\hat{i}\rangle &= |i\rangle \\ |\hat{j}\rangle &= |j\rangle, & |\hat{j}\rangle &= |j\rangle + z|i\rangle \end{aligned}$$

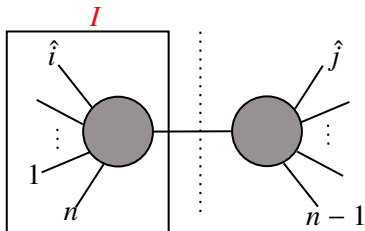
satisfying

$$k^2 = 0, \quad p_i \cdot k = 0, \quad p_j \cdot k = 0$$

For a non-trivial subset of generic momenta $\{p_i\}_{i \in I}$

$$\hat{P}_I^2 = P_I^2 - 2zP_I \cdot k = -\frac{P_I^2}{z_I}(z - z_I)$$

with $z_I = \frac{P_I^2}{2P_I \cdot k}$.



Brief explanation:

We consider amplitude A_n in terms of shifted momentum $\hat{p}_i^\mu$ instead of original real momentum.

$$A_n \longrightarrow \hat{A}_n(z)$$

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If we consider the meromorphic function $\frac{\hat{A}_n(z)}{z}$ in the complex plane. From Cauchy Theorem, we can obtain

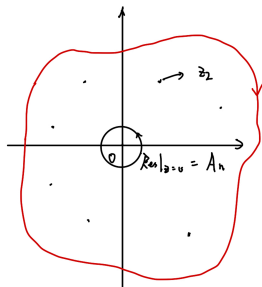
$$A_n = - \sum_{z_I} \text{Res}|_{z=z_I} \frac{\hat{A}_n(z)}{z} + B_n,$$

where B_n is the residue of the pole at $z = \infty$, called boundary term.

The most important point here is that

$$\text{Res}|_{z=0} \frac{\hat{A}_n(z)}{z} = \hat{A}_n(0) = A_n$$

It means that the original amplitude equals to the residue at origin.



$$\hat{A}_n(z) \xrightarrow{z \text{ near } z_I} \hat{A}_L(z_I) \frac{1}{\hat{P}_I^2} \hat{A}_R(z_I) = -\frac{z_I}{z - z_I} \hat{A}_L(z_I) \frac{1}{\hat{P}_I^2} \hat{A}_R(z_I)$$

This makes it easy to evaluate the residue at $z = z_I$

$$-\text{Res}|_{z=z_I} \frac{\hat{A}_n(z)}{z} = \frac{(z - z_I)z_I}{z(z - z_I)} \hat{A}_L(z_I) \frac{1}{\hat{P}_I^2} \hat{A}_R(z_I)|_{z=z_I} = \hat{A}_L(z_I) \frac{1}{\hat{P}_I^2} \hat{A}_R(z_I)$$

Large z behavior

In the BCFW formula, the boundary term B_n affects a lot

$$A_n = - \sum_{z_I} \text{Res}|_{z=z_I} \frac{\hat{A}_n(z)}{z} + B_n,$$

In most applications, one assumes or much better, proves $B_n = 0$. This is often justified by declaring a stronger condition

$$\hat{A}_n(z) \rightarrow 0 \quad \text{for} \quad z \rightarrow \infty$$

Here I show the large z behavior for gluon scattering and Einstein gravity

$[i \setminus j]$	+	-
+	$1/z$	z^3
-	$1/z$	$1/z$

$\epsilon_1 \setminus \epsilon_2$	--	-+	++
--	$1/z^2$	$1/z^2$	$1/z^2$
-+	z^2	z^2	$1/z^2$
++	z^6	z^2	$1/z^2$

proved by using soft-collinear EFT (N. Arkani-Hamed and J. Kaplan, [\[arXiv:0801.2385 \[hep-th\]\]](https://arxiv.org/abs/0801.2385)).

On-shell 3-point can be completely determined

For real momentum:

$$\begin{aligned}\langle 12 \rangle &= ([21])^* \\ \Rightarrow (p_1 + p_2)^2 = \langle 12 \rangle [21] &= 0 \quad \text{only if} \quad \langle 12 \rangle, [21] = 0\end{aligned}$$

If the momentum is **complexed**, we have

$$\begin{aligned}\langle 12 \rangle &\neq [21]^* \\ \Rightarrow (p_1 + p_2)^2 = \langle 12 \rangle [21] &= 0 \quad \Longleftrightarrow \quad \langle 12 \rangle = 0 \quad \text{or} \quad [21] = 0\end{aligned}$$

Then we can obtain

$$\mathcal{A} \propto \frac{\langle \rangle}{\langle \rangle \langle \rangle} \quad \text{or} \quad \frac{[]}{[] []}$$

It means that 3-point amplitude depends only on angle brackets or square brackets. Here I choose the first case to give an example

$$A_3(1^{h_1}, 2^{h_2}, 3^{h_3}) = c \langle 12 \rangle^{x_{12}} \langle 13 \rangle^{x_{13}} \langle 23 \rangle^{x_{23}},$$

Little group scaling tells us that

$$t_1^{2h_1} A_3(1^{h_1}, 2^{h_2}, 3^{h_3}) = c t_1^{-x_{12}} t_1^{-x_{13}} \langle 12 \rangle^{x_{12}} \langle 13 \rangle^{x_{13}} \langle 23 \rangle^{x_{23}}.$$

We can obtain

$$2h_1 = -x_{12} - x_{13}$$

Similarly, we can also obtain

$$2h_2 = -x_{12} - x_{23}, \quad 2h_3 = -x_{13} - x_{23}.$$

Then all index can be solved from this system of equations, so that

$A_3^{h_1 h_2 h_3} = c \langle 12 \rangle^{h_3 - h_1 - h_2} \langle 31 \rangle^{h_2 - h_1 - h_3} \langle 23 \rangle^{h_1 - h_2 - h_3}$	$h_1 + h_2 + h_3 < 0$
$A_3^{h_1 h_2 h_3} = c' [12]^{h_1 + h_2 - h_3} [23]^{h_2 + h_3 - h_1} [31]^{h_3 + h_1 - h_2}$	$h_1 + h_2 + h_3 > 0$

★ All massless on-shell 3-point amplitudes are completely determined by little group scaling!

Example: 3-garviton amplitude

$$\mathcal{M}_3(1^- 2^- 3^+) = \frac{\langle 12 \rangle^6}{\langle 23 \rangle^2 \langle 31 \rangle^2} = \left(\frac{\langle 12 \rangle^3}{\langle 23 \rangle \langle 31 \rangle} \right)^2 = A_3[1^- 2^- 3^+]^2$$

When does it work?

Of course, BCFW does not only apply to pure Yang-Mills

- Pure Yang-Mills theory.
- Scalar - QED and QCD
- Scalar theory $\lambda\phi^4$.
- $\mathcal{N} = 4$ super Yang - Mills theory.
- Einstein Gravity and some modified gravity.
- $\mathcal{N} = 8$ Supergravity.
-

So far: Foundations

- Reviewed the structure of **BCFW recursion relation**
- Applied to:
 - **Pure Yang-Mills** theory
 - Tree-level MHV amplitudes
 - Color-ordered partial amplitudes

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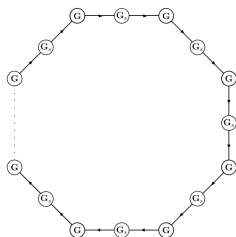
Next: Realistic Models

- Move beyond massless gauge theory
- Consider:
 - **(De)constructed gauge theories**
- Key questions:
 - Can BCFW still apply?
 - What new structures emerge?

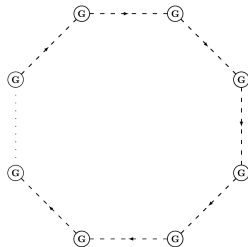
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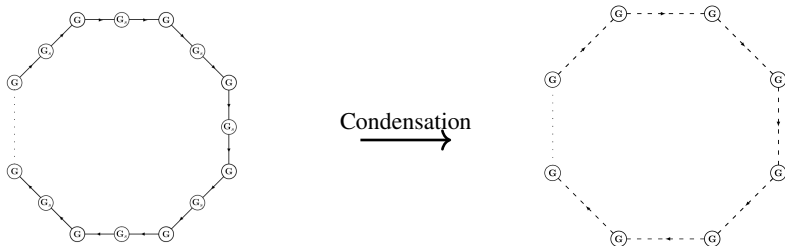
Introduction of (De)Constructed gauge theory



Condensation
 $\longrightarrow$



Introduction of (De)Constructed gauge theory



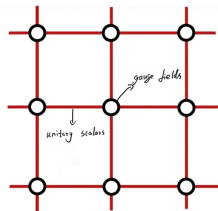
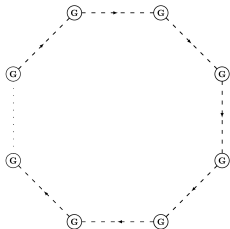
The Lagrangian can be written like

$$\mathcal{L} = - \sum_{i=1}^N \frac{1}{2} \text{Tr}(F_i^2) + \sum_{i=1}^N \text{Tr} \left[(D_\mu \Phi_i)^\dagger (D^\mu \Phi_i) \right],$$

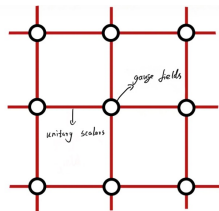
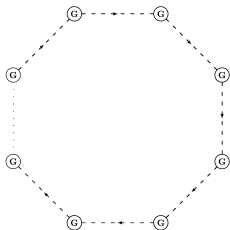
here F_i refers to the i th gauge field strength. The scalar field Φ_i transforms under the **bi-fundamental** representation, and the covariant derivative equals to

$$D_\mu \Phi_i = \partial_\mu \Phi_i - ig_i A_{i\mu} \Phi_i + ig_{i+1} \Phi_i A_{i+1\mu}.$$

It has been proposed that this model actually discretized a **five-dimension gauge theory** with gauge group $SU(m)$, where only the fifth dimension are latticed. So it is an effective theory for 5d gauge theory.



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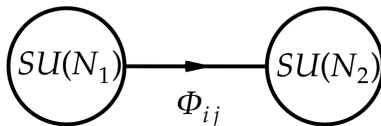


After higgsing the scalar field, we can obtain a spectrum

$$M_k^2 = 4g^2 f_s^2 \sin^2 \left(\frac{\pi k}{N} \right)$$

This is precisely the **Kaluza-Klein** spectrum under S^1 compactification.

Amplitudes from BCFW



$$\mathcal{L} = -\frac{1}{2}\text{Tr}(F_1)^2 - \frac{1}{2}\text{Tr}(F_2)^2 + \text{Tr}[(D_\mu\Phi)^\dagger(D^\mu\Phi)],$$

From the Lagrangian, we have known that there are only two kinds of 3 point amplitude (+, - : helicity $\Phi, \Phi^\dagger$: charge of scalar)

$$A[1^\Phi 2^{\Phi^\dagger} 3^+] = \frac{[23][31]}{[12]}, \quad A[1^\Phi 2^{\Phi^\dagger} 3^-] = \frac{\langle 23 \rangle \langle 31 \rangle}{\langle 12 \rangle}$$
$$A[3^+ 4^+ 5^-] = \frac{[34]^3}{[45][53]}, \quad A[3^- 4^- 5^+] = \frac{\langle 34 \rangle^3}{\langle 45 \rangle \langle 53 \rangle}$$

By using the 3 point building block, we can construct 4 point colorordered amplitudes from BCFW recursion relation.

- nV_1 or nV_2

This part is completely the same as the pure gluon amplitude, so we can directly borrow the existing results.

$$\text{Parke - Talyor Formula : } A[\cdots i^- \cdots j^- \cdots] = \frac{\langle ij \rangle^4}{\langle 12 \rangle \langle 23 \rangle \cdots \langle n1 \rangle}$$

Notice that this formula only applies to MHV amplitudes, although the NMHV can be completely solved.

The color factor in this sector looks like

$$(T^{a_1} T^{a_2} \dots T^{a_n})_{ij}$$

so what we need to notice is just the order of gauge boson.

The amplitudes can be computed like

- $\Phi^\dagger V_1 V_1 \Phi$

$$A[1^\Phi 2^{\Phi^\dagger} 3^+ 4^-] = (-1) \frac{\langle 14 \rangle^2 \langle 24 \rangle^2}{\langle 12 \rangle \langle 23 \rangle \langle 34 \rangle \langle 41 \rangle} \quad (\text{Parke -Talyor like Formula})$$

- $\Phi^\dagger V_1 V_1 V_1 \Phi$

$$A[1^\Phi 2^{\Phi^\dagger} 3^+ 4^+ 5^-] = \frac{\langle 15 \rangle^2 \langle 25 \rangle^2}{\langle 12 \rangle \langle 23 \rangle \langle 34 \rangle \langle 45 \rangle \langle 51 \rangle}$$

- $\Phi^\dagger(nV_1)\Phi$

$$A[1^\Phi 2^{\Phi^\dagger} \cdots (n+2)^-] = (-1)^{n+1} \frac{\langle 1, n+2 \rangle^2 \langle 2, n+2 \rangle^2}{\langle 12 \rangle \langle 23 \rangle \cdots \langle n+1, n+2 \rangle \langle n+2, 1 \rangle}$$

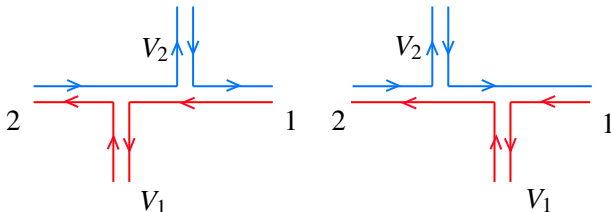
★ Bonus relation:

$$A[1^\Phi 2^{\Phi^\dagger} 3^+ 4^+] = 0 \quad \Rightarrow \quad A[1^\Phi 2^{\Phi^\dagger} 3^+ \cdots n^+] = 0$$

For the amplitude $\Phi(nV_2)\Phi^\dagger$, we can obtain nearly the same expression.

Pure 2-site amplitude

The straightforward way to observe the color structure in this case is double line notation as follows



The color factor here have special form like

$$(T_1^{a_1} T_1^{a_2} \cdots T_1^{a_{n_1}})_{ij} (T_2^{b_1} T_2^{b_2} \cdots T_2^{b_{n_2}})_{\bar{j}\bar{i}}$$

we can notice that the relative order between two gauge group do not affect the color structure, but the order inside the gauge group matters.

So we introduce the **OPP (Order Preserving Permutation)**, for example

$$\begin{array}{ccc} (3_1^+, 4_1^+, 5_2^+, 6_2^-) & (3_1^+, 5_2^+, 4_1^+, 6_2^-) & (3_1^+, 5_2^+, 6_2^-, 4_1^+) \\ (5_2^+, 3_1^+, 4_1^+, 6_2^-) & (5_2^+, 3_1^+, 6_2^-, 4_1^+) & (5_2^+, 6_2^-, 3_1^+, 4_1^+) \end{array}$$

represent the same order under OPP.

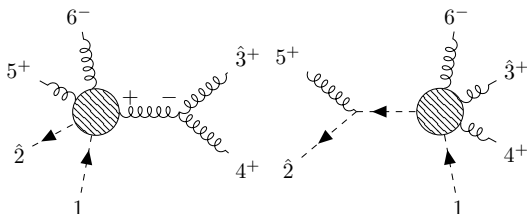
- $\Phi V_2 \Phi^\dagger V_1$

$$A[1^\Phi 2^{\Phi^\dagger} 3_1^+ 4_2^-] = \frac{\langle 14 \rangle \langle 24 \rangle}{\langle 13 \rangle \langle 23 \rangle}$$

- $\Phi V_2 \Phi^\dagger V_1 V_1$

$$A[1^\Phi 2^{\Phi^\dagger} 3_1^+ 4_1^+ 5_2^-] = (-1) \frac{\langle 25 \rangle^2 \langle 15 \rangle^2}{\langle 23 \rangle \langle 34 \rangle \langle 41 \rangle \langle 25 \rangle \langle 51 \rangle}$$

Here I show the concrete computation process



$$\begin{aligned}
 A_1 &= \frac{(-1)\langle\hat{2}6\rangle^2\langle16\rangle^2}{\langle25\rangle\langle56\rangle\langle61\rangle\langle\hat{2}\hat{I}\rangle\langle\hat{I}1\rangle} \times \frac{1}{s_{34}} \times \frac{[\hat{3}4]^3}{[4\hat{I}][\hat{I}\hat{3}]} \\
 &= \frac{\langle26\rangle^2\langle16\rangle}{\langle25\rangle\langle56\rangle\langle\hat{2}\hat{I}\rangle\langle\hat{I}1\rangle} \times \frac{1}{s_{34}} \times \frac{[34]^3}{[4\hat{I}][\hat{I}\hat{3}]} \\
 &= \frac{\langle26\rangle^2\langle16\rangle\cancel{[34]^3\langle42\rangle}}{\langle25\rangle\langle56\rangle\langle41\rangle\langle32\rangle\langle43\rangle\cancel{[43][43][34]\langle24\rangle}} \\
 &= \frac{\langle26\rangle^2\langle16\rangle^2}{\langle23\rangle\langle34\rangle\langle41\rangle\langle25\rangle\langle56\rangle\langle61\rangle}
 \end{aligned}$$

- Compact formula for general case

$$A = \frac{\langle 2a \rangle^2 \langle 1a \rangle^2}{\underbrace{\langle 2\star \rangle \cdots \langle \star 1 \rangle}_{SU(N_1)} \underbrace{\langle 2* \rangle \cdots \langle *1 \rangle}_{SU(N_2)}}$$

Green: Particle with – helicity

Blue: Particle belongs to the first gauge group

Red: Particle belongs to the second gauge group

★: Order for gauge group 1

*****: Order for gauge group 2

- Compact formula for general case

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Green: Particle with – helicity

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For example, if we want to compute $A[1^\Phi 2^{\Phi^\dagger} 5_1^+ 3_1^+ 4_1^- 7_2^+ 6_2^+ 8_2^+]$:

$$A = \frac{\langle 24 \rangle^2 \langle 14 \rangle^2}{\langle 25 \rangle \langle 53 \rangle \langle 34 \rangle \langle 41 \rangle \langle 27 \rangle \langle 76 \rangle \langle 68 \rangle \langle 81 \rangle}$$

If you use Feynman diagrams, it may take several days to accomplish the computation.

Conjecture about large-z behavior

From the 6-point amplitude

$$A[1^{\Phi}2^{\Phi^{\dagger}}3_1^{+}4_1^{+}5_2^{+}6_2^{-}] = \frac{\langle 26 \rangle^2 \langle 16 \rangle^2}{\langle 23 \rangle \langle 34 \rangle \langle 41 \rangle \langle 25 \rangle \langle 56 \rangle \langle 61 \rangle}$$

we can find that if we choose

$$[2, a^{-}] \quad \text{shift} : \begin{cases} 1/z & \text{if 2 and a adjacent} \\ 1/z^2 & \text{if 2 and a not adjacent} \end{cases}$$

similar to different shift

$$[a^{-}, 2] \quad \text{shift} : \begin{cases} 1/z & \text{if 2 and a adjacent} \\ 1/z^2 & \text{if 2 and a not adjacent} \end{cases}$$

From the structure of amplitudes in two site model, it is suggestive to suspect the large-z in n site model

$$[1(\text{or } 2), a^{-}] \quad \text{shift} : \begin{cases} 1/z^{n-1} & \text{if 2 and a adjacent} \\ 1/z^n & \text{if 2 and a not adjacent} \end{cases}$$

Contents

- 1 Motivation
- 2 Preliminary
- 3 Model and Computation
- 4 Some problems and extends**
- 5 Summary

How about NMHV?

First, let us review the NMHV amplitudes for gluon scattering.

Still we begin with the simplest case – **Split-helicity** NMHV like $A_6[1^- 2^- 3^- 4^+ 5^+ 6^+]$.

Here we choose $[1, 2\rangle$ shift

$$A_6[1^- 2^- 3^- 4^+ 5^+ 6^+] = \text{diagram A} + \text{diagram B}$$

Diagram A: A t-channel exchange between legs 1 and 2. The left vertex has legs 1⁻ and 6⁺. The right vertex has legs 2⁻, 3⁻, 4⁺, and 5⁺. The internal lines are labeled $\hat{P}_{16}$ with a minus sign on the left and a plus sign on the right.

Diagram B: A t-channel exchange between legs 5 and 6. The left vertex has legs 1⁻ and 5⁺. The right vertex has legs 2⁻, 3⁻, and 4⁺. The internal lines are labeled $\hat{P}_{156}$ with a minus sign on the left and a plus sign on the right.

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Here we choose $[1, 2\rangle$ shift

$$A_6[1^- 2^- 3^- 4^+ 5^+ 6^+] = \text{diagram A} + \text{diagram B}$$

Diagram A: A t-channel exchange between legs (1, 6) and (2, 3, 4, 5). The propagator is labeled $\hat{P}_{16}$. The external legs are labeled $\hat{1}^-$, 6^+ , $\hat{2}^-$, 3^- , 4^+ , and 5^+ . The internal lines are labeled $-$ and $+$.

Diagram B: A u-channel exchange between legs (1, 5) and (2, 3, 4, 6). The propagator is labeled $\hat{P}_{156}$. The external legs are labeled $\hat{1}^-$, 5^+ , $\hat{2}^-$, 3^- , 4^+ , and 6^+ . The internal lines are labeled $-$ and $+$.

- Diagram B includes a propagator $1/P_{156}^2$, so there is a 3-particle pole $P_{156}^2 = 0$. But by inspecting the external order, it seems that there's no difference between $(- + +)$ channel 561 and 345. We should expect the amplitude to have a pole also at $P_{345}^2 = 0$.

$$\text{diagram A} = \frac{\langle \hat{1} \hat{P}_{16} \rangle^3}{\langle \hat{P}_{16} 6 \rangle \langle 6 \hat{1} \rangle} \times \frac{1}{P_{16}^2} \times \frac{\langle \hat{2} 3 \rangle^3}{\langle 3 4 \rangle \langle 4 5 \rangle \langle 5 \hat{P}_{16} \rangle \langle \hat{P}_{16} 2 \rangle}$$

$$\langle \hat{2}\hat{P}_{16} \rangle [\hat{P}_{16}3] = \langle 21 \rangle [\hat{1}3] + \langle \hat{2}6 \rangle [63]$$

It follows from $\hat{P}_{16}^2 = 0$ that $z_{16} = -\frac{[16]}{[26]}$, so

$$\langle \hat{2}\hat{P}_{16} \rangle [\hat{P}_{16}3] = -\frac{[36]}{[26]} (\langle 12 \rangle [12] + \langle 16 \rangle [16] + \langle 26 \rangle [26]) = -\frac{[36]}{[26]} P_{126}^2$$

The 3-particle pole P_{126}^2 is encoded inside the BCFW channel !

- Full expression

$$A_6[1^- 2^- 3^- 4^+ 5^+ 6^+] = \frac{\langle 3|1+2|6 \rangle^3}{P_{126}^2 [21][16] \langle 34 \rangle \langle 45 \rangle \langle 5|1+6|2 \rangle} + \frac{\langle 1|5+6|4 \rangle^3}{P_{156}^2 [23][34] \langle 56 \rangle \langle 61 \rangle \langle 5|1+6|2 \rangle}.$$

The factor $\langle 5|1+6|2 \rangle$ does not correspond to a physical pole of the scattering amplitude: it is a **spurious pole**.

There has been interesting paper investigating how to systematically cancel the spurious poles, like "A. Hodges, JHEP 1305, 35 (2013) [arXiv:0905.1473 [hep-th]]."

- We utilize the $[1, 2\rangle$ shift before, what happens if we change to $[2, 1\rangle$ shift?

$$A_6[1^- 2^- 3^- 4^+ 5^+ 6^+] = \text{diagram A'} + \text{diagram B'} + \text{diagram C'}$$

diagram A' anti-MHV $\times$ NMHV

diagram B' MHV $\times$ MHV

diagram C' MHV $\times$ MHV

diagram A' = anti-MHV $\times$ NMHV, as opposed to diagram A = MHV $\times$ MHV.

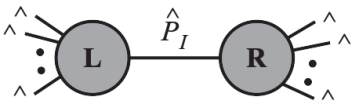
The equivalence between two different shift is related to powerful residue theorem (N. Arkani-Hamed, F. Cachazo, C. Cheung, and J. Kaplan, [arXiv:0907.5418 [hep-th]].) and Grassmannian.

CSW(Cachazo-Svrcek-Witten) expansion

We can also consider a shift that is implemented via a “holomorphic” square-spinor shift:

$$|\hat{i}\rangle = |i\rangle + zc_i |X], \quad |\hat{i}\rangle = |i\rangle$$

Here $|X]$ is an arbitrary reference spinor and the coefficients c_i satisfy $\sum_{i=1}^n c_i |i\rangle = 0$.

$$A_n^{\text{NMHV}} = \sum_{\text{diagrams } I} \text{Diagram } I$$
A Feynman diagram representing a term in the CSW expansion. It consists of two circular vertices, labeled 'L' and 'R', connected by a horizontal internal line. The internal line is labeled with a momentum vector $\hat{P}_I$ above it. Vertex 'L' has three external lines extending to the left, each ending in a small triangle pointing left. Vertex 'R' has three external lines extending to the right, each ending in a small triangle pointing right. The external lines are labeled with indices 1, 2, and 3 above them.

There are two possibilities: anti-MHV₃(= 0) × NMHV or MHV × MHV.

For example, the 6pt split NMHV amplitude

$$\begin{aligned}
 A_n[1^- 2^- 3^- 4^+ 5^+ 6^+] = & \text{Diagram 1} + \text{Diagram 2} + \text{Diagram 3} + \text{Diagram 4} + \text{Diagram 5} + \text{Diagram 6} \\
 & + \text{Diagram 7} + \text{Diagram 8} + \text{Diagram 9} \\
 & = \frac{\langle 1\hat{P}_I \rangle^4}{\langle 1\hat{P}_I \rangle \langle \hat{P}_I 5 \rangle \langle 56 \rangle \langle 61 \rangle} \frac{1}{P_{156}^2} \frac{\langle 23 \rangle^4}{\langle 23 \rangle \langle 34 \rangle \langle 4\hat{P}_I \rangle \langle \hat{P}_I 2 \rangle} .
 \end{aligned}$$

The diagrams are 3-point and 4-point vertices connected by internal lines, with external legs labeled with momenta and helicities. The vertices are represented by black circles, and the internal lines are labeled with '+' or '-' signs.

For example, the 6pt split NMHV amplitude

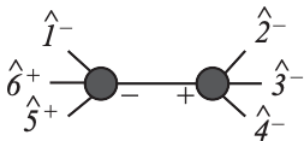
$$\begin{aligned}
 A_n[1^- 2^- 3^- 4^+ 5^+ 6^+] = & \text{Diagram 1} + \text{Diagram 2} + \text{Diagram 3} \\
 & + \text{Diagram 4} + \text{Diagram 5} + \text{Diagram 6} \\
 & = \frac{\langle 1\hat{P}_I \rangle^4}{\langle 1\hat{P}_I \rangle \langle \hat{P}_I 5 \rangle \langle 56 \rangle \langle 61 \rangle} \frac{1}{P_{156}^2} \frac{\langle 23 \rangle^4}{\langle 23 \rangle \langle 34 \rangle \langle 4\hat{P}_I \rangle \langle \hat{P}_I 2 \rangle} .
 \end{aligned}$$

We can write

$$|\hat{P}_I\rangle \frac{[\hat{P}_I X]}{[\hat{P}_I X]} = \hat{P}_I |X\rangle \frac{1}{[\hat{P}_I X]} = P_I |X\rangle \frac{1}{[\hat{P}_I X]}$$

We can use the prescription

$$|\hat{P}_I\rangle \rightarrow P_I|X]$$



$$= \frac{\langle 1|P_{156}|X\rangle^4}{\langle 1|P_{156}|X\rangle\langle 5|P_{156}|X\rangle\langle 56\rangle\langle 61\rangle} \cdot \frac{1}{P_{156}^2} \cdot \frac{\langle 23\rangle^4}{\langle 23\rangle\langle 34\rangle\langle 4|P_{156}|X\rangle\langle 2|P_{156}|X\rangle}.$$

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- 1 Motivation
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- Introduce the on-shell method, including BCFW recursion relation, spinor-helicity formalism, etc.
- Introduce a (de)constructed gauge theory model, which is an effective field theory for 5 dimension gauge theory.
- Much of the scattering amplitudes in this model can be recursively computed by BCFW, and some compact formulas are offered.

Thanks for your attention!

Helicity

Helicity is defined as the projection of a particle's spin vector $\vec{S}$ onto the direction of its momentum $\vec{p}$:

$$h = \frac{\vec{S} \cdot \vec{p}}{|\vec{p}|}$$

Spinor-Helicity Formalism

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$$h = \frac{\vec{S} \cdot \vec{p}}{|\vec{p}|}$$

S-matrix is a function of momentum p_i and helicity h_i

$$\mathcal{M}(p_i, h_i)$$

How can we catch the information of helicity?

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Massless Case:

- Momenta in spinor form:

$$p_\mu \sigma^\mu = p_{\alpha\dot{\alpha}} = p_\alpha \tilde{p}_{\dot{\alpha}} = |p\rangle[p|$$

Large z behavior

In the BCFW formula, the boundary term B_n affects a lot

$$A_n = - \sum_{z_I} \text{Res}|_{z=z_I} \frac{\hat{A}_n(z)}{z} + B_n,$$

In most applications, one assumes or much better, proves $B_n = 0$. This is often justified by declaring a stronger condition

$$\hat{A}_n(z) \rightarrow 0 \quad \text{for} \quad z \rightarrow \infty$$

Here I show the large z behavior for gluon scattering

$[i \setminus j]$	+	-
+	$1/z$	z^3
-	$1/z$	$1/z$

On-shell 3-point for real momentum

Because of the constrain from momentum conservation and on-shell condition

$$p_1 = \kappa p_3, \quad p_2 = (1 - \kappa)p_3 \quad (\text{Collinear})$$

All of the contribution

$$(p_1 \cdot p_2), \quad (p_1 \cdot p_3), \quad (p_2 \cdot p_3) = 0$$

In terms of Spinor- Helicity variable, we have

$$2p_1 \cdot p_2 = \langle 12 \rangle [21] = 0 \longrightarrow \langle 12 \rangle = [21]^* = 0$$

We can not obtain any thing nontrivial from 3-point!

Of course, you can introduce non-minimal interaction

$$\mathcal{L}_3 \ni \frac{1}{\Lambda^2} \bar{\Psi} \not{D} (\square \Psi)$$

but it still equals to 0 under the on-shell condition.

On-shell 3-point can be completely determined

For the complex momentum, we have

$$|1\rangle \propto |2\rangle \propto |3\rangle \quad \text{or} \quad |1] \propto |2] \propto |3]$$

$$A_3^{h_1 h_2 h_3} = c \langle 12 \rangle^{h_3 - h_1 - h_2} \langle 31 \rangle^{h_2 - h_1 - h_3} \langle 23 \rangle^{h_1 - h_2 - h_3} \quad h_1 + h_2 + h_3 < 0$$

$$A_3^{h_1 h_2 h_3} = c' [12]^{h_1 + h_2 - h_3} [23]^{h_2 + h_3 - h_1} [31]^{h_3 + h_1 - h_2} \quad h_1 + h_2 + h_3 > 0$$

★ **All massless on-shell 3-point amplitudes are completely determined by little group scaling!**

Example: 3-gluon amplitude

$$A_3(g_1^-, g_2^-, g_3^+) = g \frac{\langle 12 \rangle^3}{\langle 23 \rangle \langle 31 \rangle}$$

Scattering Amplitudes from BCFW

For simplicity, we start from the two-site gauge theory with gauge fields V_1, V_2 and scalar fields $\Phi, \Phi^\dagger$.

$$\mathcal{L} = -\frac{1}{2}\text{Tr}(F_1)^2 - \frac{1}{2}\text{Tr}(F_2)^2 + \text{Tr}[(D_\mu\Phi)^\dagger(D^\mu\Phi)],$$

We only focus on the following amplitudes:

$$nV_1, \quad nV_2, \quad \Phi^\dagger nV_1 \Phi, \quad \Phi nV_2 \Phi^\dagger, \quad \Phi^\dagger \Phi \Phi^\dagger \Phi$$

here n can be any positive integer.

More specifically, it helps us to prove P. T. formula

$$\begin{array}{ccccc}
 3\text{pt.} & \longrightarrow & 4\text{pt.} & \longrightarrow & 5\text{pt.} & \longrightarrow & \cdots \\
 \frac{\langle 12 \rangle^4}{\langle 12 \rangle \langle 23 \rangle \langle 31 \rangle} & & \frac{\langle 12 \rangle^4}{\langle 12 \rangle \langle 23 \rangle \langle 34 \rangle \langle 41 \rangle} & & \frac{\langle 12 \rangle^4}{\langle 12 \rangle \langle 23 \rangle \langle 34 \rangle \langle 45 \rangle \langle 51 \rangle} & &
 \end{array}$$

More specifically, it helps us to prove P. T. formula

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 \frac{\langle 12 \rangle^4}{\langle 12 \rangle \langle 23 \rangle \langle 31 \rangle} \quad \frac{\langle 12 \rangle^4}{\langle 12 \rangle \langle 23 \rangle \langle 34 \rangle \langle 41 \rangle} \quad \frac{\langle 12 \rangle^4}{\langle 12 \rangle \langle 23 \rangle \langle 34 \rangle \langle 45 \rangle \langle 51 \rangle}
 \end{array}$$

$$\Rightarrow: \quad A[1^+ \cdots i^- (i+1)^+ \cdots j^- (j+1)^+ \cdots n^+] = \frac{\langle ij \rangle^4}{\langle 12 \rangle \langle 23 \rangle \cdots \langle n1 \rangle}$$

★ This is the power of BCFW recursion relation.